

K-Nearest Neighbor Algorithm

Dataset preparation:

Randomly split the dataset into Training (70%), Validation (15%), and

Test (15%) `set Train_set=[], Val_set=[], Test_set=[]`

//Following code shuffles your dataset list

1. for each sample S in the dataset:
2. generate a random number R in the range of [0,1]
3. if $R \geq 0$ and $R \leq 0.7$
4. append S in Train_set
5. elif $R > 0.7$ and $R \leq 0.85$
6. append S in Val_set
7. else:
8. append S in Test_set

100 samples
70 training
15 validation
15 test

KNN Classification:

Use Iris data [iris](#),

K = 5

1. for each sample V in the VALIDATION set:
2. for each sample T in the TRAINING set:
3. Find Euclidean distance between V_x (features $\rightarrow N-1$) and T_x (features $\rightarrow N-1$)
4. Store T and the distance in list L
5. Sort L in ascending order of distance
6. Take the first K samples
7. Take the majority class from the K samples (this is the detected class for sample V)
8. Now, check if this class is correct or not

9. Calculate $\text{validation_accuracy} = (\text{correct VALIDATION samples}) / (\text{total VALIDATION samples}) * 100$

- Similarly, calculate validation accuracy for $K = 1, 3, 5, 10, 15$
- Make a table with 2 columns: K and Validation Accuracy ([report template](#))
- Now, take the K with the **highest** Validation Accuracy
- Use this best K to determine **Test Accuracy** (Simply replace the VALIDATION set with the TEST set)

KNN Regression:

Use diabetes data [diabetes](#)

Error = 0

1. for each sample V in the VALIDATION set:
 2. for each sample T in the TRAINING set:
 3. Find Euclidean distance between Vx and Tx
 4. Store Tx and the distance in list L
 5. Sort L in ascending order
 6. Take the first K samples
 7. Take the average output of the K samples (this is the determined output for sample V)
 8. $\text{Error} = \text{Error} + (\text{V true output} - \text{V determined output})^2$
 9. Calculate $\text{Mean_Squared_Error} = \text{Error} / (\text{total number of samples in VALIDATION set})$
- Similarly, calculate Mean_Squared_Error for K = 1, 3, 5, 10, 15
 - Make a table with 2 columns: K and **Mean_Squared_Error** ([report template](#))
 - Now, take the K with the **minimum** Mean_Squared_Error
 - Use this best K to determine **Mean_Squared_Error for the Test set** (Simply replace the VALIDATION set with TEST set.

Instruction

- Submit the .ipynb file and a report ([report template](#)) .pdf file.
- **DO NOT USE LIBRARIES SUCH AS: "Sklearn", "Scikit learning" or "pandas" for this assignment.**